

Intraoperative and postoperative anaesthetic and analgesic effect of multipoint transcutaneous electrical acupuncture stimulation combined with sufentanil anaesthesia in patients undergoing supratentorial craniotomy

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ABSTRACT

Objectives To investigate the anaesthetic and analgesic effect of multipoint transcutaneous electrical acupuncture stimulation (TEAS) during supratentorial tumour resection for postoperative recovery and side effects.

Methods In a blinded clinical trial, 92 patients scheduled for supratentorial craniotomy under general anaesthesia were randomly allocated into either a multipoint TEAS (n=46) or a sham TEAS group (n=46). All patients received total intravenous anaesthesia (TIVA) with propofol and sufentanil. The target concentration of sufentanil was adjusted and recorded according to mean arterial pressure (MAP), heart rate (HR) and bispectral index (BIS). Patients in the TEAS group received TEAS 30 min before anaesthesia induction and this was maintained throughout the operation at four pairs of acupuncture points. Postoperative pain, recovery and side effects were evaluated.

Results Eighty-eight patients completed the study. Continuous monitoring of MAP, HR and BIS showed stable values with no significant differences between the two groups ($p>0.05$). Sufentanil target plasma concentration in TEAS patients was significantly lower at some time points during supratentorial craniotomy, and total sufentanil consumption was significantly higher in the sham group ($p<0.05$). Postoperative recovery and pain were significantly improved by TEAS ($p<0.001$), without the postoperative side effects.

Conclusions Multipoint TEAS at both proximal and distal points combined with TIVA can significantly decrease intraoperative sufentanil

requirements, increase pain relief on postoperative day 1 and improve postoperative recovery of patients during supratentorial tumour resection, with no significant increase of side effects. These findings suggest that multipoint TEAS may be clinically effective as an adjunct to analgesia in intraoperative anaesthesia and postoperative pain treatment and may speed recovery.

Trial registration number Chinese Clinical Trial Registry (registration number ChiCTR-TRC-10001078).

INTRODUCTION

Acupuncture is a traditional Chinese medicine and has been extensively used as a non-pharmacological analgesic therapy since being developed 2500 years ago. Although opioids are widely used as analgesics in general anaesthesia, the unpleasant side effects and delayed postoperative recovery have set some limitations on their clinical use.¹ Several clinical studies have shown that acupuncture may be effective in improving postoperative analgesia, reducing intraoperative anaesthetic requirement and immunosuppression, and reducing the incidence of anaesthesia-related side effects.^{2–5} Stimulation of specific points, by needles or electrodes, releases neurochemical substrates that may block the incoming pain information.⁶ However, several studies also found that acupuncture and related techniques did not significantly reduce pain during the operation,⁷ and not only



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failed to reduce, but actually increased, anaesthetic requirements.⁸ These conflicting results may be partially explained by the different acupuncture points and exogenous anaesthetic used, though other factors, such as the timing of the intervention and the patients' level of consciousness, may also be involved.

Transcutaneous electric acupuncture stimulation (TEAS), or 'needleless acupuncture', is a low-skill alternative to needle-based electroacupuncture (EA) that combines the advantages of both acupuncture and transcutaneous electrical nerve stimulation by using electrode pads placed on the acupuncture points instead of piercing the skin with needles.⁹ Compared with acupuncture or EA, TEAS is a non-invasive technique with very low risk of infection and it can potentially be applied by any anaesthetist or preoperative personnel with minimal training, increasing the reproducibility of acupuncture-like stimulation. Several lines of evidence indicate that TEAS can significantly improve the quality of recovery and decrease the incidence of anaesthesia-related side effects for patients undergoing surgery.¹⁰ However, most clinical studies have focused on the postoperative pain relief and side effects of TEAS and, to our knowledge, its intraoperative effects have not been well characterised.

Our previous studies showed that EA at both proximal and distal (peripheral) acupuncture points showed beneficial intraoperative and postoperative effects on patients undergoing supratentorial tumour resection.¹¹ EA combined with general anaesthesia significantly shortened the anaesthesia recovery time and decreased both the sevoflurane requirement and postoperative side effects during resection of a supratentorial tumour.¹² TEAS, a therapeutic approach has some similarities to EA,¹³ and may also achieve the same effect as EA in intraoperative anaesthesia, postoperative analgesia and in the reduction of side effects.

This study was designed to investigate the intraoperative and postoperative anaesthetic effects of multi-point TEAS at both proximal and distal (peripheral) points during supratentorial craniotomy, when in combination with sufentanil anaesthesia, including its effects on intraoperative analgesic requirements, postoperative pain relief, recovery and improvement in side effects.

PATIENTS AND METHODS

Patients

This randomised, blind, controlled trial was approved by the ethics review board of Beijing Tiantan Hospital and registered in the Chinese Clinical Trial Registry (registration number ChiCTR-TRC-10001078). All participants gave their written informed consent before being enrolled in this study. A total of 92 patients aged 18–60 years, of American Society of Anesthesiologists physical status I/II, scheduled for supratentorial craniotomy, were eligible for this trial between January 2008 and December 2011. Exclusion

criteria were as follows: women in pregnancy or lactation; patients with the complication of severe respiratory and circulatory system diseases; long-term heavy smokers; patients with body mass index $>35 \text{ kg/m}^2$; emergency patients. Patients with operation time $>8 \text{ h}$ and operative blood loss $>2500 \text{ mL}$ were withdrawn from the study. Ninety-two patients meeting the criteria were recruited and randomised, but four patients were withdrawn from the final analysis.

Protocol

Consecutive patients were enrolled and randomly allocated into the TEAS group or sham group according to a random number table generated by a computer. No premedication (analgesics and tranquiliser) was given to any of the patients. In the operation room, peripheral intravenous infusion was started and non-invasive arterial blood pressure, heart rate (HR), pulse oximetry saturation, bispectral index (BIS) and exhalation gas were monitored.

Transcutaneous electrical acupuncture stimulation

Patients in the TEAS group received preoperative TEAS starting 30 min before anaesthesia induction, maintained throughout the operation and terminated at the end of surgery (removal of headstock). Four pairs of electrodes ($3 \times 3 \text{ cm}^2$) were applied at five pairs of acupuncture points: *Hegu* (LI4) and *Waiguan* (TE5), *Jinmen* (BL63) and *Taichong* (LR3), *Zusanli* (ST36) and *Qixu* (GB 40), and *Fengchi* (GB20) with *Tianzhu* (BL10) and *Cuanzhu* (BL2) with *Yuyao* (EX-HN4) on the craniotomy side. GB20 and BL10, BL2 and EX-HN4 were adjacent and covered by the same electrode, respectively. TEAS was performed using a Han's acupoint nerve stimulator (LH202H, Beijing Huawei Co, Ltd, Beijing, China) with a dense-disperse frequency of 2/100 Hz (alternated once every 3 s; 0.6 ms at 2 Hz and 0.2 ms at 100 Hz). The intensity of stimulation was set at 4.89 ± 2.15 , 6.79 ± 3.51 , 7.04 ± 3.35 and 5.61 ± 2.13 , respectively, according to the maximal tolerance of patients and maintained throughout the operation. In the sham group, the gel electrodes were applied in the same therapeutic setting to ensure blinding of the patients, which has previously proved to be successful.¹⁴ The patients were connected to the instrument, but electronic stimulation was not applied. All the acupuncture-related operations were performed by the physicians of Beijing University of Chinese Medicine. Only the acupuncturist was informed of the randomisation allocation using opaque, sealed envelopes, just before the onset of TEAS. Others, including the anaesthesiologists, surgeons, recovery staff, assessors and participants, were blinded to the group allocation. Postoperative data were collected by researchers who were also blinded to the study design.

Anaesthesia

Anaesthesia was induced by intravenous infusion of propofol (1% Diprivan) and sufentanil ($0.2 \mu\text{g/mL}$)

to maintain their target plasma concentrations at 5 µg/mL and 0.5 ng/mL, respectively. Sufentanil was administered as target-controlled infusion using the pharmacokinetic model of Bovill *et al.*¹⁵ After loss of consciousness, target plasma concentrations of propofol and sufentanil were each subsequently reduced to 3.2 µg/mL and 0.3 ng/mL. Vecuronium bromide 0.1 mg/kg was given intravenously to facilitate tracheal intubation and muscle relaxation. Mechanical ventilation with a 40% oxygen/air mixture was applied with the tidal volume of 10 mL/kg, respiratory frequency of 12 times/min and gas flow of 1 L/min set.

The dose of propofol was stable during the operation for anaesthetic maintenance and sufentanil concentration was adjusted to maintain the mean arterial pressure (MAP), HR and BIS in the basic range of +10% to -20%. Vecuronium bromide of 0.05 mg/kg was injected intravenously and intermittently to retain muscle relaxation (T4/T1<25%) according to the results of muscle relaxation (train of four stimulation). Local infiltration anaesthesia of the incision was obtained by 0.5% ropivacaine.

Postoperative analgesia

After the operation, analgesic compound liquid, consisting of 50 mL of normal saline and 100 µg of sufentanil, 1 mL/h was given through a sustaining pump with a single injecting volume of 0.5 mL and locking time of 10 min. Patients were asked to rate the degree of pain on a Visual Analogue Scale (VAS) consisting of a numerical scale from 0 to 10 points; with 0 being 'no pain' and 10 being 'maximum pain'.¹⁰

Measurements

The demographic characteristics of the patients were recorded, as well as the intensity and duration of TEAS. MAP, HR, BIS and sufentanil target plasma concentrations were recorded at different stages of the operation, including preoperation stage (from loss of

consciousness to before scalp cutting), craniotomy stage (from scalp cutting to dura cutting), intracranial stage (from after dura opening to before dura stitching) and cranial closure stage (from dura stitching to complete scalp stitching).

End points

The primary outcome of this study was the consumption of anaesthetics. The secondary end points were the time to spontaneous respiration, extubation time, eye-opening time, time to spontaneous movement, time to reorientation, and time to discharge from the operating room. After recovery room admission, the postoperative side effects, including incidence of respiratory depression, nausea, vomiting and pain, were also recorded at postoperative days 1, 2 and 3.

Statistical analysis

All data input and the statistical analysis were performed in the department of epidemiology and hygienic statistics of Capital Medical University, Beijing, China, using SAS V9.0 software (SAS Institute Inc, Cary, North Carolina, USA). The measurement data with normal distribution are expressed as mean ±SD and were analysed by a two sample t test. Discrete data were analysed by the χ^2 test or Fisher's exact probability test, as appropriate. A p value of <0.05 was considered statistically significant.

RESULTS

Patient characteristics and surgical details

Ninety-two patients were enrolled in the study; four patients (4.3%), two in each group, were withdrawn because they met the withdrawal criteria. Two were excluded owing to an operation time of >8 h (TEAS group patient: 16 h; sham group patient: 10 h) and two were excluded owing to operative blood loss >2500 mL (TEAS group patient: 3800 mL; sham

Table 1 Characteristics and surgical details of patients receiving TEAS or sham treatment

	Sham group (n=44)	TEAS group (n=44)	p Value
Age (years)	43.8±11.47	42.1±10.83	0.486
Weight (kg)	65.2±10.02	65.4±13.35	0.90
Gender (M/F)	16/28	15/29	0.594
ASA I/II	26/18	24/20	0.487
Operation time (min)	255.8±73.02	235.9±62.57	0.19
Anaesthesia time (min)	304.9±84.48	287.5±71.02	0.29
Type of operation (cases)			0.184
Meningioma	18	18	
Glioma	18	20	
Metastatic tumours or others	8	6	
Infusion amount (mL)	3466.3±739.84	3348.0±884.50	0.693
Estimated blood loss (mL)	557.9±408.28	446.0±518.32	0.144
Urine amount (mL)	1507.5±752.36	1414.0±703.86	0.594

Results are shown as mean±SD unless stated otherwise.

ASA I/II, American Society of Anesthesiologists physical status I/II; TEAS, transcutaneous electrical acupuncture stimulation.

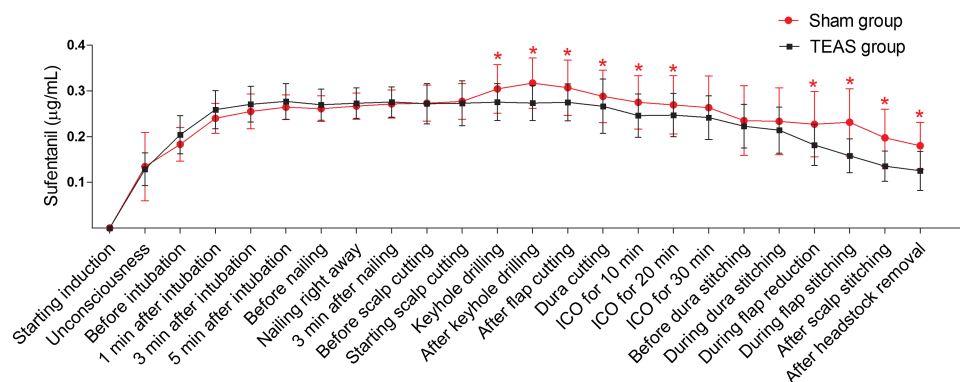


Figure 1 Continuous monitoring of sufentanil target plasma concentration during anaesthetic induction and maintenance of supratentorial tumour resection in patients receiving TEAS or sham treatment. ICO, intracranial operation; TEAS, transcutaneous electrical acupuncture stimulation; * $p < 0.05$ versus the sham group.

group patient: 5000 mL). The data of 88 patients (31 male, 57 female) were analysed. The characteristics of patients, and the type and durations of procedures were not significantly different in the groups receiving TEAS or sham treatment (table 1).

MAP, HR and BIS were continuously monitored during anaesthesia induction and maintenance of the patients undergoing supratentorial tumour resection. MAP and HR remained stable during anaesthesia induction and maintenance, and there were no statistical significant differences at each time point between the TEAS and sham groups (see online supplementary figures S1 and S2; $p > 0.05$). Continuous monitoring of BIS showed values between 40 and 60 after anaesthesia induction and maintenance of anaesthesia, and no significant difference was seen between patients receiving TEAS or sham treatment (see online supplementary figure S3; $p > 0.05$).

Sufentanil target plasma concentration in patients receiving TEAS or sham treatment

There was no significant difference in sufentanil target plasma concentration at each time point before keyhole drilling and from 20 min intracranial operation to dura stitching, while, during keyhole drilling to 10 min intracranial operation and from flap

reduction to the end of operation, target plasma concentration of sufentanil was significantly lower in TEAS patients than in sham-treated ones (figure 1, $p < 0.05$).

Intraoperative anaesthetics consumption and clinical recovery from anaesthesia

There were significant differences between the two groups in the total dose of sufentanil consumption, the time to spontaneous respiration, extubation time and the time to discharge from the operating room, all favouring the TEAS group (table 2). The total dose of propofol given, eye-opening time, the time to spontaneous movement and the time to reorientation were also found to be reduced or shortened, although differences did not reach statistical significance.

Postoperative pain and side effects

No respiratory depression was noted in either group and the incidences of nausea and vomiting did not differ significantly between the two groups (table 3). Pain scores were significantly lower at postoperative day 1 in the TEAS group than in the sham group, whereas the VAS score in the TEAS group was comparatively higher at postoperative days 2 and 3 (figure 2).

Table 2 Intraoperative anaesthetics consumption and clinical recovery from anaesthesia in patients receiving TEAS or sham treatment

	Sham group (n=44)	TEAS group (n=44)	p Value
Sufentanil (µg)	117.7±37.95	95.6±21.76*	0.02
Propofol (mg)	234.1±71.30	216.3±67.72	0.245
Spontaneous respiration (min)	23.4±14.13	16.0±9.99**	0.005
Extubation (min)	33.7±17.40	26.6±14.17*	0.036
Eye-opening (min)	28.2±17.74	22.2±12.04	0.066
Spontaneous movement (min)	34.3±22.59	27.5±13.79	0.092
Reorientation (min)	42.7±19.80	35.3±16.11	0.076
Operating room discharge (min)	48.7±16.70	39.8±15.74*	0.013

Results are shown as mean±SD.

* $p < 0.05$, ** $p < 0.01$ versus the sham group.

TEAS, transcutaneous electrical acupuncture stimulation.

Table 3 Postoperative side effects in patients receiving TEAS or sham treatment

	Sham group (n=44)			TEAS group (n=44)		
	Day 1	Day 2	Day 3	Day 1	Day 2	Day 3
Nausea	9 (20.5)	5 (11.4)	2 (4.5)	11 (25.0)	2 (4.5)	2 (4.5)
Vomiting	8 (18.2)	4 (9.1)	0 (0.0)	9 (20.5)	2 (4.5)	1 (2.3)

Results are shown as number (%).

TEAS, transcutaneous electrical acupuncture stimulation.

DISCUSSION

As a widespread complementary therapy, acupuncture appears to be effective in the treatment of many symptoms and/or disorders, partly by regulating the functions of the autonomic nervous system and neuroendocrine system,⁹ especially in improving postoperative pain relief and reducing postoperative side effects.¹⁶ Given the inherent side effects and limitations of conventional opioid analgesics, increasing interest has been focused on the use of acupuncture or acupuncture-related techniques in combination with general anaesthesia, as well as their effect on intraoperative anaesthetic and opioid requirements.¹⁷ TEAS is an acupuncture-like technique that may produce effects similar to those elicited by acupuncture or EA treatment.¹⁴ In this study, TEAS was used in coordination with sufentanil to observe the intraoperative and postoperative effects of TEAS on patients undergoing supratentorial tumour resection. The results showed that either sufentanil combined with TEAS or sufentanil alone produced surgical anaesthesia for supratentorial craniotomy. Sufentanil requirement, postoperative recovery time and degree of pain on postoperative day 1 were significantly lower in patients treated by TEAS than in those without TEAS ($p < 0.05$ or $p < 0.01$), illustrating clear adjuvant analgesic effects. In contrast, the VAS pain score was higher in the TEAS group on postoperative days 2 and 3. The explanation for this is unknown and may relate to the short-term anaesthetic effects of TEAS,

so that the analgesic effect of TEAS may stop after the operation without further stimulation.¹⁸ However, the postoperative side effects, like vomiting, nausea and respiratory depression, were not significantly different between the two groups.

A number of studies have provided evidence that acupuncture can induce neural responses in different brain areas.^{19–20} BL63, LR3, ST36 and GB40 are well-known analgesia acupuncture points generally used for pain relief during craniotomy,²¹ and stimulation at points LI4 and TE5 has also been used for postoperative pain relief during clinical surgery.^{10–14} Corroborating the results of these studies, our previous studies showed evidence suggesting that EA at these acupuncture points combined with sevoflurane anaesthesia during supratentorial tumour resection significantly reduced the sevoflurane required and the postoperative side effects and improved the postoperative recovery from anaesthesia.^{12–22} EA at both the aforementioned distal acupuncture points and the proximal acupuncture points on the craniotomy side, including GB20, BL10, BL2 and EX-HN4, could reduce the anaesthetic requirement and improve the postoperative recovery of patients undergoing supratentorial craniotomy. Alternating stimulation with low and high frequency (2 and 100 Hz) can stimulate the release of the three intrinsic opioid peptides (endorphin, enkephalin and dynorphin) in the central nervous system to produce a general analgesic action.¹³ Therefore, in this research, four pairs of acupuncture points were selected for TEAS treatment.

In our study, propofol was administered for anaesthesia maintenance using a target-controlled infusion system and sufentanil administration was adjusted according to the arterial pressure, HR and BIS, a well-established method of monitoring anaesthetic depth and consequently a useful tool in assessing the risk of awareness during induction and maintenance of anaesthesia.²³ Our results showed that the values of MAP, HR and BIS remained stable during induction and maintenance of anaesthesia, with an adequate level of anaesthesia obtained (BIS was maintained between 40 and 60 during all operations). There was no significant difference between the TEAS and sham groups, and surgical details of patients, like operation time, anaesthesia time, type of operation, infusion amount, urine amount, estimated blood loss, were not significantly different, indicating that either the TEAS-combined with sufentanil or sufentanil alone as administered in our study provided a clinically optimal condition for craniotomy. Although the MAP, HR and BIS values were not statistically significantly different, MAP in the TEAS group was slightly lower during surgery, while HR showed no consistent pattern, BIS was slightly higher during part of the surgical time, unlike the results reported by Si *et al.*²⁴ The potential mechanisms remain unknown. The propofol concentration remained consistent during

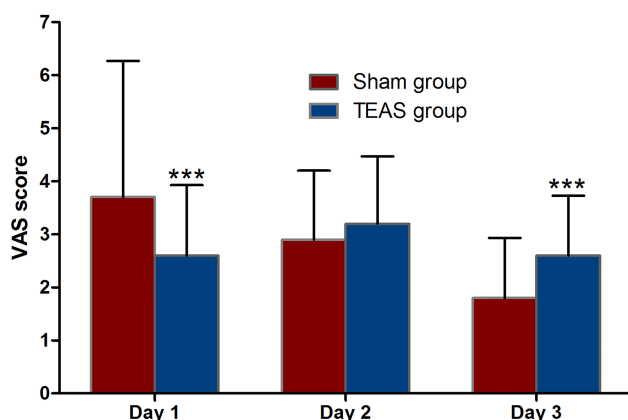


Figure 2 Postoperative VAS scores of patients receiving TEAS or sham treatment. TEAS, transcutaneous electrical acupuncture stimulation; VAS, Visual Analogue Scale. *** $p < 0.001$ versus the sham group.

surgery and the consumption of propofol was similar in the two groups. Shafer and Varvel²⁵ reported that the IC50 value of sufentanil for electroencephalography depression was 0.69 ng/mL, suggesting that the TEAS treatment might explain the lower MAP and HR and higher BIS seen in our study, although more evidence is needed.

Our present study showed that the sufentanil target plasma concentration was significantly higher in the sham-treated group during supratentorial craniotomy surgery and TEAS treatment can help reduce the intraoperative sufentanil requirement. Meanwhile, the time to spontaneous respiration, extubation time and the time to discharge from the operating room were also found to favour the patient receiving TEAS treatment, in agreement with the results of previous studies.^{12–14} The VAS scores were also found to be reduced by TEAS treatment during postoperative day 1. However, unlike previous studies, which indicated that TEAS is an effective method for prophylaxis of postoperative nausea and vomiting,¹⁴ we found no statistically significant difference in the incidence and severity of postoperative complications, such as respiratory depression, nausea and vomiting. Similar results were obtained in recent study by Wang *et al.*,¹⁴ who hypothesised that this lack of effect may be attributable to the low incidence of postoperative nausea and vomiting after intravenous anaesthesia, or the antiemetics they used. This may not be the case in our study, where no antiemetic was given, and instead the different acupuncture points and intensity of TEAS used may have a role.^{26–27}

In summary, the results of this blinded controlled study indicated that multipoint TEAS at both proximal and distal acupuncture points combined with sufentanil anaesthesia can significantly reduce the intraoperative analgesic requirements, strengthen pain relief on the first postoperative day, and improve the postoperative recovery of patients undergoing supratentorial tumour resection, although the postoperative side effects were not found to be significantly improved. These findings suggest that multipoint TEAS may be clinically effective as an adjunct to sufentanil in intraoperative anaesthesia and postoperative analgesia and recovery.

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